

Rinku: A Distributed Ledger For Mesh-Native Systems

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Abstract

Existing distributed ledgers assume persistent, well-connected infrastructure. They halt when the network partitions, require always-on RPC endpoints for state verification, and their smart contracts presume a synchronized world. These assumptions break in intermittently connected environments - ad-hoc mesh networks, field deployments with unreliable backhaul, maritime and remote operations, disaster response coordination, and autonomous agent fleets - where connectivity gaps are routine operating conditions, not exceptional failures.

Rinku is a DAG-based distributed ledger built around three primitives designed for these environments. **Tunable consistency** navigates the CAP tradeoff dynamically: delivering checkpoint finality when quorum is reachable, degrading gracefully to provisional availability during partitions, and deterministically reconciling divergent state when connectivity is restored. **VerifiableObjects** are self-contained, URL-encoded cryptographic proofs that carry everything needed for offline verification - no full node, no RPC endpoint, no network access. Any participant can receive a payment proof, verify it locally with a single BLS check, and act on it before ever touching external infrastructure. **BYOP (Bring Your Own Proof)** smart contracts accept VerifiableObjects as inputs, enabling contract logic to execute against proven external state without synchronous cross-chain or cross-contract calls - composable without centralized coordination.

Together, these primitives describe a ledger designed for environments where partitions are a first-class operating condition. The native RKU token has a hard cap of 30 million units with checkpoint-based emission and weighted proof-of-stake rewards.

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1. Introduction

Distributed ledgers have achieved remarkable success in environments where network connectivity is reliable and persistent. Bitcoin's Nakamoto consensus and Ethereum's Gasper protocol deliver strong probabilistic or deterministic finality under the assumption that a sufficient fraction of participants can communicate within reasonable time. BFT-family protocols - Tendermint, HotStuff, Bullshark, Mysticeti - push finality latency down to seconds or sub-seconds, but all share a fundamental constraint: they halt when the network cannot reach quorum.

This constraint is acceptable for data-center-class infrastructure where partitions are rare and short-lived. It is insufficient for a growing class of environments where intermittent connectivity is the norm: ad-hoc wireless mesh networks, mobile-first deployments in regions with unreliable backhaul, maritime and remote field operations, disaster response coordination, autonomous agent fleets, and industrial IoT meshes. These environments require the same economic primitives - verifiable payments, enforceable contracts, trustless state sharing - that well-connected infrastructure readily supports.

The distributed systems literature offers rich solutions for partition tolerance - CRDTs, eventual consistency, Bayou-style conflict resolution, anti-entropy protocols - but these techniques have seen limited adoption in production blockchain design. The gap exists because financial state is not naturally commutative: transferring tokens is an inherently non-idempotent operation where order matters and conflicts have economic consequences.

Rinku bridges this gap. It provides a distributed ledger designed for environments where partitions are a first-class operating condition, not a side-effect to be circumnavigated. The protocol maintains strong finality when the network is connected and degrades gracefully to provisional operation during partitions, with a deterministic merge protocol that reconciles state when connectivity is restored.

1.1 The Partition Problem

Traditional blockchains prioritize a single canonical history with strong finality but sacrifice availability - the network halts when partitions prevent quorum. There are typically 3 categories for how these halts are handled:

- Category 1 (Nakamoto): Live during partition, reorg losers on reconnect, with no intelligent reconciliation
- Category 2 (BFT): Halt finalization, queue in mempool; safety over liveness
- Category 3 (Hybrid/Ethereum): Tips keep moving, finality pauses; halfway measure

So far, these have proven fairly sufficient for strongly connected infrastructure, but for networks that are disjointed or regularly fragmented there still remains to be a more appropriate solution.

1.2 Rinku's Position

Rinku takes a different position: **tunable consistency**. When the network is fully connected, Rinku Convergent Consensus (RCC) delivers sub-second transaction finality through leaderless stake-weighted convergence, with periodic state snapshots for sync and pruning. When partitions occur, the network degrades gracefully to provisional acceptance - transactions continue locally, and state reconciles deterministically when partitions heal.

The core invariant: **no honest user is prevented from transacting during a partition**. Naturally, the cost of this guarantee is that some transactions may be rolled back during merge if they conflict with transactions from other partitions. Intentional abuse is economically penalized.

1.3 VerifiableObjects as the User-Facing Primitive

Rinku's most distinctive user-facing concept is the **VerifiableObject** (VO): a self-contained, URL-encoded cryptographic proof that carries all data necessary for offline verification. Every meaningful output of the protocol - a payment confirmation, a contract execution receipt, a trust score attestation - is expressible as a portable `rinku://vo/` URL. This collapses the distinction between "querying the chain" and "holding a proof" - any party with a VO can verify its claim without a full node, an RPC endpoint, or any network access at all. Section 3 expands on this design much further.

2. Trust Model & Terminology

Before describing the protocol, we establish the trust assumptions and define key terms precisely. These definitions apply throughout the paper.

2.1 Genesis Trust Anchor

The Rinku network is bootstrapped from a **genesis checkpoint** signed by an initial validator set defined in node configuration (**GENESIS_VALIDATORS**). This genesis checkpoint establishes:

- The initial account state (genesis allocation of 6,000,000 RKU)
- The founding validator set with initial stakes
- The root of the checkpoint hash chain

All subsequent state is cryptographically derived from this anchor. Non-genesis nodes verify the genesis validator set against their own configuration and will wipe and resync if the genesis state is inconsistent.

2.2 Validator Set Evolution

The validator set evolves through staking and unstaking transactions on the ledger:

Joining the validator set. Any account may become a validator by submitting a **Stake** transaction locking at least 100 RKU. Upon finalization (inclusion in a checkpoint signed by the current quorum), the staking account is added to the active validator set maintained by the **ValidatorIdentityService**. The new validator's BLS public key is registered and becomes eligible for checkpoint voting at the next height.

Leaving the validator set. A validator submits an **Unstake** transaction. The stake enters a 24-hour cooldown period during which it remains locked and slashable but no longer earns rewards. After cooldown, the stake is returned to the liquid balance. A 14-day unbonding window applies for slashing purposes - evidence of misbehavior during the bonded period can still trigger slashing during unbonding.

Checkpoint signing. Each checkpoint is signed by the validators active at that height. The **ConsensusService** maintains a frozen snapshot of the validator set for each voting round, ensuring that staking changes mid-round do not affect quorum calculation. New validators participate in signing starting from the checkpoint following their stake finalization.

Genesis-to-runtime transition. The genesis node creates the initial validator accounts and registers their stakes. Non-genesis nodes bootstrap by syncing the genesis checkpoint and adopting the genesis validator set from **GENESIS_VALIDATORS** configuration. During sync, if a node detects that its local genesis state is inconsistent with the peer's, it performs a full database wipe and resync to ensure all nodes share the same trust root. The runtime validator set evolves independently from the genesis set as staking transactions are processed.

2.3 Terminology: Acceptance vs. Finality

This paper uses precise terminology for transaction lifecycle states:

Term	Meaning	Guarantee
Submitted	Transaction received by a node and inserted into the DAG	No durability guarantee
Accepted	Transaction has received convergence confirmation from $>2/3$ active stake	High confidence of inclusion; state applied immediately; durable via snapshot
Finalized	Transaction referenced by a checkpoint signed by $>2/3$ total stake quorum	Irreversible under honest majority assumption
Provisionally finalized	Transaction referenced by a provisional checkpoint during partition mode	Valid within the local partition; subject to rollback during merge
Reconciled	Previously provisional transaction that survived the merge protocol	Equivalent to finalized; referenced by the merge checkpoint
Rolled back	Previously provisional transaction that was rejected during merge	Removed from canonical state; kept in DAG for auditability if cascade victim

2.4 What Receipts Prove

Receipt Type	Mode	Anchor	Proves
Convergence ACK	Normal	Validator stake votes	$>2/3$ stake has seen and accepted the transaction. State is applied immediately upon convergence.
Checkpoint receipt	Normal	BLS-signed checkpoint with state root	Transaction is included in a finalized

Receipt Type	Mode	Anchor	Proves
			checkpoint. Irreversible.
Provisional receipt	Partition	Provisional checkpoint with <code>partition_epoch</code>	Transaction is locally finalized within this partition. Subject to rollback.
Reconciliation receipt	Post-merge	Merge checkpoint with <code>merge_report_hash</code>	Transaction survived reconciliation and is now globally finalized.
Rollback receipt	Post-merge	Merge report	Transaction was rolled back, with reason (conflict loser, cascade victim).

2.5 Threat Model

Rinku's security properties rely on the following assumptions:

Honest majority. The protocol assumes that validators controlling more than 2/3 of total staked RKU follow the protocol honestly. This is the standard BFT assumption. Under this assumption, finalized checkpoints are irreversible - a conflicting checkpoint would require >1/3 stake to sign two different checkpoint hashes at the same height, which is detected and slashed (section 12).

Byzantine fault tolerance. Up to $f < n/3$ validators (by stake weight) may behave arbitrarily - equivocating, withholding votes, or broadcasting invalid messages. The protocol guarantees safety (no conflicting finalized checkpoints) as long as the honest majority assumption holds. Liveness requires >2/3 stake to be reachable for checkpoint finality; during partitions, liveness is maintained locally through provisional checkpoints (Section 8).

Partition attacker model. An adversary capable of partitioning the network can:

- Cause some partitions to operate under provisional finality (reduced but functional)
- Cause some transactions to be rolled back during merge if they conflict across partitions
- Trigger cascade rollbacks that affect innocent users whose transactions depended on rolled-back funds

An adversary **cannot**:

- Forge transactions (ECDSA P-256 signatures)
- Cause conflicting finalized (non-provisional) checkpoints (requires >2/3 honest stake)
- Double-spend without detection and penalty (nonce reuse is detected during merge; Section 12)
- Exploit partition tolerance for profit without incurring economic penalties that exceed the potential gain (Section 12.4)

Sybil resistance. Identity and influence in the protocol are derived from staked RKU. Weight calculations use sub-linear stake scaling ($\text{stake}^{0.5} * 2.0$) to reduce the advantage of large stakers while maintaining Sybil resistance. Note: sub-linear scaling technically *rewards* stake splitting — N validators at stake S/N each yield aggregate weight proportional to $S^{0.5} * N^{0.5}$, which grows with N. Sybil resistance is therefore load-bearing on the minimum stake requirement ($\text{MIN_STAKE} = 100$ RKU): creating N Sybil identities requires $N \times 100$ RKU locked capital, limiting the practical splitting advantage. See Section 12.5 for detailed analysis.

3. VerifiableObject System

VerifiableObjects (VOs) are Rinku's universal container for portable, self-proving cryptographic claims. Every proof type in the system produces a `rinku://vo/` URL with embedded proof data and freshness metadata. VOs are the primary interface between the Rinku protocol and external consumers - they are how the ledger communicates provable facts to the world. By embedding all verification data within the proof itself, VOs eliminate the requirement for full-node infrastructure, RPC endpoints, or reliable network connectivity for state verification.

3.1 Proof Types

Type	Description	Use Case
ContractOutput	StatefulReceipt with view keys, pre/post state roots, events	Stateless dApp verification
AccountProof	Balance, nonce, stake at a specific checkpoint	Account state verification
TxFinality	Transaction inclusion proof with Merkle path and BLS signature	Payment confirmation
WeightProof	Aggregate stake weight attestation	Trust scoring, anti-disinformation

Type	Description	Use Case
BatchProof	Multi-receipt verification with shared checkpoint context	Bulk verification
StateWitness	Sparse Merkle multiproof for contract storage keys	Stateless contract reads
Custom	Schema-defined proofs for application-specific claims	Extensibility

3.2 URL Encoding

VOs are serialized to JSON, DEFLATE-compressed, and encoded as URL-safe Base64:

`rinku://vo/<base64_compressed_json>`

Additional URI schemes for specific proof types:

- `rinku://sp/` - Self-contained proofs (account state with full Merkle path to checkpoint)
- `rinku://asp/` - Account state proofs (compact)

The URL encoding is designed so that a VO can be shared as a hyperlink, embedded in a QR code, or passed as a transaction parameter - collapsing the boundary between "data" and "proof of data."

3.3 Proof Freshness

Every VO carries optional **ProofFreshness** metadata:

- `generated_at_checkpoint` - checkpoint height at proof generation
- `generated_at_timestamp` - wall-clock time of generation
- `max_age_checkpoints` - optional expiry window

Verifiers can enforce proof age limits, preventing the use of stale proofs in time-sensitive operations. This is critical for BYOP transactions (Section 3.4) where a contract must ensure it is acting on recent state.

3.4 BYOP (Bring Your Own Proof) Transactions

Contracts accept **ProofInput** arrays as transaction parameters. Each proof input carries a **VerifiableObject** and a **ProofExpectation** specifying:

- Required proof type

- Chain ID (for potential cross-chain use)
- Minimum checkpoint height
- Expected state root

The runtime validates all proofs before contract execution begins and injects verified data into the WASM context under the `proof.<label>.<key>` namespace. This enables contracts to act as their own oracles - consuming proven facts from other contracts, accounts, or even external chains without synchronous state access.

Security against proof replay. Proof freshness requirements mitigate replay attacks where a valid but stale proof is resubmitted to a contract. The `ProofExpectation` includes a `max_age_checkpoints` field; the runtime computes the proof's age as `current_checkpoint_height - generated_at_checkpoint` and rejects proofs that exceed the caller's specified age limit. This enforcement occurs at the verifier's constraint level, not the proof's own `max_age_checkpoints` - ensuring that the consumer of a proof, not its producer, controls freshness requirements.

Cross-contract receipt composition. A contract can consume the `StatefulReceipt` of another contract's execution as a `ProofInput`. For example, a lending contract can accept a price oracle contract's receipt as proof of the current collateral value, without making a synchronous cross-contract call. The lending contract validates the oracle receipt's Merkle proof against the state root, checks freshness, and proceeds with the proven value. This pattern enables composability without re-entrancy risk.

4. Design Philosophy

4.1 URL-Native State

Rinku embeds the entire ledger state within cryptographically-linked URLs. Every proof, receipt, and state witness can be encoded as a self-contained URL that carries all data necessary for offline verification. This eliminates the dependency on full-node infrastructure for trust - any party holding a Rinku URL can independently verify its claim against a checkpoint anchor.

4.2 Self-Provable Ledger

Rinku's proof architecture differs fundamentally from SPV (Simplified Payment Verification) and traditional light client models:

SPV proofs (Bitcoin) prove transaction inclusion in a block via Merkle path but require the verifier to trust block headers, which in turn requires following the longest-chain rule. The verifier

must either maintain a header chain or trust a third party that does. SPV proofs do not prove account state - only transaction inclusion.

Light clients (Ethereum Beacon Chain, Cosmos IBC) verify state transitions by tracking a committee of validators and their signatures. They require an ongoing connection to at least one honest full node to follow the validator set evolution. Without this connection, a light client's view becomes stale and unverifiable.

Rinku self-contained proofs carry everything needed for offline verification in a single URL:

1. The claim itself (account balance, transaction finality, contract output)
2. A Merkle inclusion proof connecting the claim to a checkpoint state root
3. The BLS aggregate signature over the checkpoint hash
4. A signer bitmap identifying which validators signed
5. Signer membership proofs (Merkle Sum Tree) proving each signer was a member of the validator set with their claimed stake weight

A verifier holding a self-contained proof and a trusted checkpoint (or the genesis checkpoint) can verify the claim without any network access, any RPC endpoint, or any trust in the proof's provider. The proof is self-authenticating - either the math checks out or it doesn't.

This removes the reliance of "blockchain infrastructure" for read operations. Where Ethereum requires archive nodes, RPC providers (Infura, Alchemy), and client libraries to answer "what is Alice's balance?", Rinku answers the same question with a URL that anyone can verify with a quick, single cryptographic check.

4.3 Tunable Consistency

Rinku does not claim to "solve" or "beat" the CAP theorem. Instead, it navigates CAP by dynamically selecting the appropriate tradeoff:

- **Normal operation:** CP-like strong finality via checkpoint quorum. Transactions achieve irreversible settlement through BLS-signed checkpoints backed by $>2/3$ total stake. This is not classical linearizability in the distributed systems sense, but provides the practical guarantees users expect: once finalized, a transaction cannot be reverted.
- **Partition mode:** Provisional availability. Transactions continue locally with explicitly labeled provisional finality. Clients are informed that these confirmations are subject to rollback.
- **Post-partition:** Deterministic convergence to global consistency through the merge protocol (Section 9). All nodes independently compute identical reconciled state from the same inputs.

The innovation is not the mode switching itself - it is the **transaction taxonomy and reconciliation semantics** that make this mode switching practical for financial state without unacceptable rollback risk.

4.4 CAP Analysis

Formally, Rinku's position in the CAP design space is:

Mode	Consistency	Availability	Partition Tolerance
Normal	Strong (convergence + finality)	Full (convergence + snapshot)	CP: consistency is preserved; if quorum becomes unreachable, finality pauses rather than forking
Partitioned	Provisional (local consistency only)	Partial (Safe and BoundedSpend transactions only; CpOnly transactions are rejected)	AP: availability is preserved for eligible transaction types at the cost of global consistency
Post-merge	Strong (deterministic reconciliation)	Temporarily reduced (merge computation)	Transitional: partition has healed; deterministic convergence restores CP semantics

Note: partition tolerance is a design property of the protocol, not a runtime state. Rinku is always partition-tolerant in the sense that partitions do not cause data loss or protocol-level inconsistency - the system's *response* to partitions is what varies between CP (Normal) and AP (Partitioned) behavior.

During partition, Rinku explicitly sacrifices global consistency in favor of local availability for eligible transaction types. The key insight is that this sacrifice is **bounded and recoverable**: the transaction taxonomy (Section 9.8) limits which operations can proceed, the partition budget system bounds the economic exposure, and the merge protocol deterministically recovers global consistency.

This is not eventual consistency in the CRDT sense - Rinku does not guarantee that all operations commute. Instead, it guarantees that non-commutative operations (financial

transfers) are either merge-safe by design (within partition budget) or subject to explicit, deterministic conflict resolution with graduated economic penalties for abuse.

5. Network Architecture

5.1 Peer-to-Peer Layer

Rinku uses pure libp2p with gossipsub for all inter-node communication. There are no HTTP-based node-to-node calls. The protocol operates on two channels:

- **Gossip topic (`rinku/1.0.0`):** Transaction broadcast, checkpoint announcements, tip announcements, merge payloads
- **Request-response protocol (`/rinku/sync/1.0.0`):** Checkpoint sync, delta sync, snapshot recovery, presync, partition visibility queries

Note: 1.0.0 reflects current protocol version.

5.2 Node Roles

Validator nodes participate in consensus by staking RKU, producing convergence ACKs for transaction finality, and proposing state snapshots when elected as proposer. Validators run the full protocol stack including the partition detector, merge orchestrator, and tip consolidation service. Only the genesis node creates the initial validator accounts and registers their stakes; non-genesis validators adopt the validator set during sync.

Full nodes maintain a complete copy of the DAG and state but do not participate in checkpoint voting. They validate all transactions and checkpoints, serve API requests, relay gossip messages, and can independently verify the entire chain from genesis. Full nodes contribute to network resilience by increasing the number of honest relays.

Light clients (TBA..?). The self-contained proof architecture (Section 4.2) enables a lightweight verification mode where clients hold no ledger state. A light client can verify any claim by receiving a `rinku://sp/` URL and checking the BLS signature, Merkle proof, and signer membership proofs against a trusted checkpoint anchor. This model requires no ongoing connection to any full node - verification is a single, stateless cryptographic operation.

5.3 Synchronization

Rinku implements a multi-mode synchronization system optimized for different scenarios:

Snapshot sync (new nodes). A joining node requests a complete state snapshot from a peer. The snapshot includes all account balances, stakes, contract state, and the current checkpoint chain. After applying the snapshot, the node replays any transactions in the DAG that occurred after the snapshot's checkpoint height. Transactions are only marked `finalized: true` if they appear in the snapshot's `finalized_tx_hashes` or in checkpoint `finalized_tx_hashes` lists - preventing spurious finality attribution during recovery.

Delta sync (catching up). A node that has been briefly disconnected requests only the transactions and checkpoints created since its last known checkpoint. This is a lightweight operation that avoids retransmitting the full state. Delta sync also reconciles the validator identity service, ensuring the catching-up node's `total_active_stake` matches the network.

Presync (quick bootstrap). Before performing a full snapshot sync, a joining node performs a lightweight handshake to determine the peer's checkpoint height and validator set. This allows the node to estimate the sync workload and select the most appropriate sync mode.

Persistent storage. All state is persisted using `redb`, a lightweight embedded database. The persistent transaction counter (`total_transactions`) is stored in metadata alongside `gas_price`, `total_supply`, and `genesis_time`, ensuring accurate counts survive DAG pruning across restarts.

Ghost account prevention. During sync, account push-back filters out stale accounts (zero balance, zero nonce) to prevent state contamination from obsolete data. **Known edge case:** this heuristic incorrectly filters legitimate accounts that have been fully drained (sent their entire balance with no incoming transactions). Such accounts would need to receive funds again to reappear in the synced state. In practice, this affects only accounts with exactly zero balance *and* zero nonce — an account that ever transacted (`nonce > 0`) is preserved regardless of balance.

6. Consensus: Rinku Convergent Consensus (RCC)

Rinku implements a two-layer consensus architecture called Rinku Convergent Consensus (RCC). Layer 1 provides sub-second transaction finality through leaderless stake-weighted convergence. Layer 2 creates periodic state snapshots for synchronization, pruning, and proof generation.

Intellectual lineage. RCC draws on the convergence insight from the Avalanche protocol family [9] - that reliable broadcast with stake-weighted voting can achieve finality without leaders or rounds. The implementation uses explicit `TxConfirmAck` messages (point-vote accumulation) for per-transaction convergence and round-robin proposer-based state snapshots for periodic anchoring. Unlike Avalanche's repeated sub-sampled polling, RCC uses a single broadcast

round where all validators participate in parallel, achieving convergence in a single network round-trip.

6.1 Layer 1: Transaction Convergence

Transaction convergence provides sub-second **finality** through stake-weighted validator voting (observed median ~43ms, p95 ~200ms, p99 ~500ms; see Appendix C.2). This is the primary consensus layer - all transactions achieve finality here.

The convergence process:

1. **Broadcast.** When a validator receives a valid transaction (via API or gossip), it broadcasts a `TxConfirmBroadcast` message containing the full transaction, the validator's address, and its stake weight.
2. **Voting.** Every validator that receives the broadcast validates the transaction (signature, nonce, balance) and responds with a `TxConfirmAck` containing its stake weight and BLS signature over the transaction hash.
3. **Convergence.** The originating node accumulates ACKs. When accumulated stake exceeds 2/3 of total active stake (`CONVERGENCE_QUORUM_THRESHOLD`), the transaction is marked `Confirmed`.
4. **Execution.** Upon confirmation, the transaction's state effects are applied immediately - balances are updated, nonces are incremented, contract state is modified. The transaction achieves **finality** at this point.

Key properties:

- **Leaderless.** No leader election is required for transaction finality. All validators participate symmetrically in convergence voting.
- **Monotonic finality.** Once a transaction achieves 2/3 stake convergence, it is final. There is no subsequent phase that can revert it (under the honest majority assumption).
- **Immediate state application.** State is applied at convergence, not at snapshot boundaries. This enables sub-second end-to-end latency from submission to state update.
- **All transaction types eligible.** Every transaction type (transfer, stake, unstake, contract call) participates in convergence. There is no eligibility filter.
- **Nonce-ordered execution.** Convergence-confirmed transactions with future nonces (higher than the account's current nonce) are deferred and retried in (`from_address`, `nonce`) sorted order, ensuring sequential execution unblocks dependent transactions without rejection.

To prevent double-execution, convergence-executed transactions are tracked in a `convergence_executed` set and are not re-applied during snapshot creation.

6.2 Layer 2: State Snapshots

State snapshots are periodic, non-blocking anchors created by a round-robin proposer. They serve as sync points for new nodes, enable DAG pruning, and provide the checkpoint anchors used for proof generation.

Snapshot creation flow:

1. A round-robin proposer is elected for each snapshot height using the same deterministic stake-weighted mechanism described in Section 6.4.
2. The proposer collects all convergence-finalized transactions since the last snapshot from its local DAG.
3. Transactions are filtered using a `PROPAGATION_GRACE_MS` window (default 5,000ms) to ensure sufficient time for gossip propagation.
4. The proposer computes the `tx_merkle_root` over the transaction hashes and the `state_root` from the Sparse Merkle Trie.
5. The proposer creates a snapshot signed with its BLS key (single proposer signature, not quorum) and broadcasts it via a `CheckpointAnnouncement` gossip message.
6. Receivers verify the state root against their own local state (which should match, since all nodes independently apply state at convergence). State root mismatches trigger reconciliation via the existing delta sync protocol.

Key properties:

- **No vote collection.** Snapshots are created by a single proposer and do not require quorum voting. Receivers verify correctness by comparing state roots, not by collecting signatures.
- **Non-blocking.** Snapshot creation never blocks transaction processing. If the proposer is offline or slow, transactions continue to achieve finality via Layer 1 convergence.
- **Adaptive proposer recovery.** The snapshot proposer's stuck-iteration counter resets when unfinalized transactions appear in the DAG, preventing idle-period accumulation from trapping the proposer in a recovery-only mode.
- **Enabling proofs.** Snapshots provide the checkpoint anchors referenced by VerifiableObjects (Section 3), Merkle inclusion proofs, and finality certificates.
- **DAG pruning.** Transactions included in a snapshot can be pruned from the in-memory DAG, keeping the working set bounded.

6.2.1 Failure Modes

Scenario	Impact	Recovery
Minority validators offline	Convergence still achieves 2/3 if remaining stake suffices	Offline validators sync via delta protocol on reconnection
Proposer offline	No snapshot created at this height; transactions continue to finalize via convergence	Round-robin advances to next proposer; non-proposers fetch from peers after timeout
State root divergence	Receiver detects mismatch between snapshot state root and local state	Delta sync reconciliation resolves missing transactions
Network partition	Convergence cannot reach 2/3 global stake	Partition mode activates (Section 8); provisional acceptance within local partition

6.3 BLS Aggregate Signatures

Rinku uses the BLS12-381 signature scheme (via the `blst` library, `min_pk` variant) for checkpoint signing:

Signature scheme. Signatures are on the G2 group (96 bytes compressed); public keys are on G1 (48 bytes compressed). Each validator signs the checkpoint hash with their BLS secret key, producing an individual signature.

Aggregation. Individual validator signatures are aggregated into a single 96-byte aggregate signature using `AggregateSignature::aggregate`. This aggregation is additive - the aggregate signature can be verified against the aggregate of the corresponding public keys in a single pairing check, regardless of the number of signers.

Signer bitmaps. To identify which validators contributed to an aggregate signature without transmitting the full validator list, Rinku uses a compact bitfield. The `signer_bitmap` is a `Vec<u8>` where the i -th bit corresponds to the i -th validator in the deterministically sorted validator set. A set bit indicates that validator signed the checkpoint. This enables any verifier with knowledge of the validator set to reconstruct the aggregate public key and verify the aggregate signature.

Verification. Given a checkpoint hash, aggregate signature, signer bitmap, and validator set, verification proceeds: (1) parse the bitmap to identify signers, (2) compute the signer stake as the sum of effective stakes of all signers and check that $\text{signer_stake} / \text{total_stake} \geq 2/3$ (stake-weighted quorum, not signer-count quorum), (3) aggregate the signer public keys, (4) verify the aggregate signature against the aggregate public key and checkpoint hash. This is a constant-time operation regardless of the number of signers (a single pairing check). Verification is enforced at checkpoint reception — both the gossip handler (`CheckpointAnnouncement`) and the leader fallback path reject checkpoints that fail BLS verification. Checkpoints that lack an aggregate signature (e.g., during initial genesis bootstrap or sync of legacy checkpoints) are accepted with a warning log but not rejected, allowing gradual rollout of BLS enforcement.

Rogue-key mitigation. BLS aggregate signatures are vulnerable to rogue public key attacks where an adversary constructs a public key that cancels another validator's key during aggregation, allowing forgery of aggregate signatures. Rinku mitigates this by using the `min_pk` variant of BLS12-381 (public keys on G1, signatures on G2) and requiring proof-of-knowledge (PoK) of the BLS secret key at validator registration: each validator must produce a valid BLS signature over a registration message containing their address and stake transaction hash. This signature serves as a PoK and is verified before the validator's BLS public key is added to the active validator set. This follows the construction described by Boneh, Drijvers, and Neven [5] for secure multi-signature aggregation without the need for a deaggregation step.

Double-sign detection. The `ConsensusService` monitors for validators that sign two different checkpoint hashes at the same height. Double-signing triggers an immediate 15% stake slash and addition to a `slashed_validators` set, which reduces the validator's voting power in all pending rounds.

6.4 Leader Election

Rinku uses a deterministic, stake-weighted leader election mechanism based on verifiable randomness:

Randomness derivation. For each checkpoint height, a 32-byte seed is computed as:

```
randomness = SHA-256("RINKU_LEADER_ELECTION_V1" || checkpoint_height ||  
previous_checkpoint_hash)
```

Since all validators in consensus share the same `previous_checkpoint_hash` and target height, they all derive identical randomness. The seed is unpredictable before the previous checkpoint is finalized (depends on the previous checkpoint's hash) but deterministic afterward.

Stake-weighted selection. The active validator set is sorted by address (ensuring identical ordering on all nodes). The randomness is mapped to a target value in the range `[0,`

`total_stake`). The algorithm iterates through the sorted validators, accumulating stake until the cumulative sum exceeds the target. The validator that crosses the threshold is elected leader. Over time, each validator's probability of election is proportional to their stake.

Rotation. Because `checkpoint_height` is an input to the randomness, the leader changes every checkpoint. The combination of height-based rotation and stake-weighted selection ensures both fairness (proportional to stake) and unpredictability (cannot be known until the previous checkpoint finalizes).

Liveness fallback. If the elected leader fails to produce a checkpoint within a configurable timeout (`leader_timeout_ms`, default 45 seconds), a fallback mechanism activates. The `should_fallback` function uses a modified height input (`checkpoint_height + 1000000`) for randomness, effectively electing an emergency replacement leader. This ensures liveness even if the primary leader is offline. **Clock skew consideration:** different validators may trigger the fallback at slightly different wall-clock times due to clock skew, potentially resulting in a brief window where both the primary and fallback leaders are producing checkpoints simultaneously. This is safe — duplicate checkpoint proposals at the same height are resolved by the BLS quorum mechanism (only one can achieve 2/3 stake), and the other is discarded. However, clock skew increases the time to finality during leader failure by up to the skew magnitude.

6.5 Relationship Between Convergence and Snapshots

Property	Layer 1: Convergence	Layer 2: Snapshots
Latency	~43ms median, ~500ms p99	~5-10s (periodic)
Quorum	>2/3 active stake (ACK votes)	Single proposer (no quorum)
Stake basis	Active stake only (online validators)	N/A (proposer-signed)
State application	Immediate upon convergence	State already applied; snapshot records it
Durability	Durable via convergence; survives restart through snapshot sync	Persisted; provides sync anchor
Irreversibility	Irreversible under honest majority	Receivers verify against local state
Proof artifact	Convergence ACK set	BLS-signed snapshot with state/receipt roots

Active vs. total stake. Convergence quorum is measured against *active* stake (online validators). If a significant fraction of total stake is offline, convergence can confirm transactions with a lower absolute stake threshold. For example, if 40% of total stake is inactive, convergence requires only $>2/3$ of the remaining 60% = $>40\%$ of total stake. This is acceptable because convergence represents true finality under the honest majority assumption — the $2/3$ threshold ensures that no conflicting transaction can also achieve convergence. If active stake drops below $2/3$ of total stake, the partition detector (Section 8) activates and the network transitions to provisional mode.

7. DAG Structure & Transaction Ordering

7.1 Transaction DAG

Transactions in Rinku form a DAG rather than a linear chain. Each transaction references one or more parent transactions (DAG tips at the time of submission), creating a partial order that enables parallel processing.

7.2 Tip Selection

Rinku uses a **Sparse DAG Sampling** algorithm to prevent tip explosion while maintaining DAG connectivity and Sybil resistance.

MAX_SAMPLED_TIPS. The maximum number of parent references a transaction selects is bounded at 16. This prevents the DAG from growing excessively wide while maintaining sufficient connectivity for parallel validation.

Weighted selection with diversity. When the number of available tips exceeds 16, the sampling algorithm splits selection into two halves:

1. **Guaranteed selection (top 8):** The 8 tips with the highest sender account weight are always included. This ensures that well-staked, high-reputation transactions are preferentially referenced, providing Sybil resistance - an attacker flooding the network with low-weight transactions cannot crowd out legitimate tips.
2. **Random sampling (bottom 8):** The remaining 8 slots are filled by random sampling from all other available tips. This maintains DAG diversity, prevents the graph from narrowing to a single chain of high-weight transactions, and ensures that transactions from lower-weight (but honest) participants are eventually incorporated.

Tip consolidation. A background **TipConsolidator** service runs on validator nodes. When the tip count exceeds a threshold (default 100), it enters aggressive consolidation mode,

periodically creating **Consolidation** transactions that reference 16 tips at once. These anchor transactions merge divergent branches back into fewer tips, keeping the DAG's working set manageable.

Orphan parent handling. If a transaction arrives with parent references to hashes not found in the local DAG (orphan parents), the node automatically injects current known tips as parents. This ensures the transaction attaches to the main graph even if some referenced parents were pruned or never received.

7.3 DAG Weight Calculation

Transaction weight in the DAG context (distinct from WPoS reward weight in Section 11.3) determines transaction ordering priority, tip selection preference, and conflict resolution outcomes. It is computed as:

$$\text{effective_weight} = (\text{age_weight} * \text{balance_weight} + \text{stake_weight}) * (1.0 - \text{reputation_penalty})$$

Where:

- **age_weight:** Time-based component since transaction insertion
- **balance_weight:** Derived from the sender's account balance
- **stake_weight:** Sub-linear bonus from staked tokens, computed as $\text{stake}^{0.5} * 2.0$. The square-root scaling reduces the advantage of large stakers - doubling stake increases weight by only ~41%, not 100%. This provides meaningful Sybil resistance while limiting plutocratic concentration.
- **reputation_penalty:** A value in $[0.0, 1.0]$ reflecting accumulated partition-tolerance violations (see Section 12). An account with a 0.50 reputation penalty has its effective weight halved in all weight calculations.

The sub-linear stake weight is a deliberate design choice for decentralization: it makes it more capital-efficient to distribute stake across multiple honest validators than to concentrate it in a single large validator, while still ensuring that staked participants have significantly more influence than unstaked ones.

7.4 Cumulative Weight & Conflict Resolution

Intra-partition fork resolution. Within a single connected network (no partition), DAG forks are resolved by cumulative weight. The cumulative weight of a transaction is the sum of its own weight plus the weights of all transactions that directly or transitively reference it as a parent. When two transactions conflict (same sender, same nonce), the transaction with higher cumulative weight in the DAG is preferred. This mechanism provides probabilistic convergence similar to Bitcoin's longest-chain rule, but using stake-weighted attestation rather than proof-of-work.

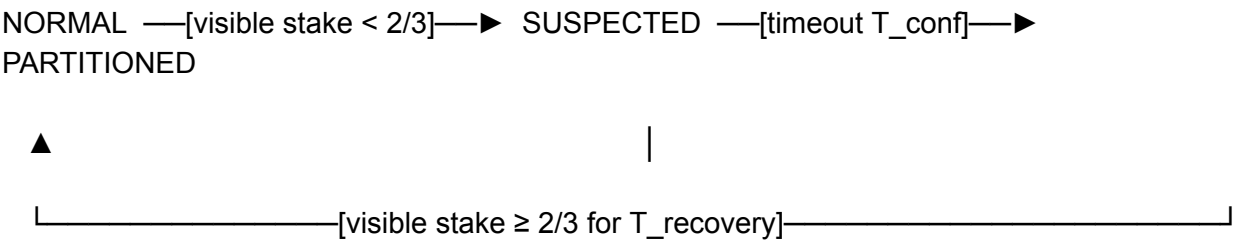
Cross-partition conflict resolution. When partitions heal and the merge protocol runs (Section 9), conflict resolution uses a more sophisticated multi-factor algorithm that considers cumulative weight, visible stake percentage, and lexicographic hash tiebreaking. This is necessary because cumulative weight alone is not a fair comparison across partitions of different sizes - a transaction in a 60%-stake partition would naturally accumulate more weight than a transaction in a 40%-stake partition, even if both are equally valid.

8. Partition Tolerance

Consider any network where sub-groups regularly lose connectivity for 30-120 seconds: a fleet of field devices on an unreliable mesh, a maritime vessel cluster with intermittent satellite uplink, or a disaster response team operating on degraded infrastructure. In these environments, partition mode is the expected operating state, not an edge case. The transaction classification system ensures that data-only operations (Safe) continue uninterrupted, bounded financial transactions (BoundedSpend) proceed within pre-configured risk limits, and consensus-critical operations like staking changes (CpOnly) wait for full network reconnection.

8.1 Detection

Rinku implements a three-state partition detector that continuously monitors network health:



Parameter	Default	Description
T_conf	30s	Confirmation timeout before declaring partition
T_recovery	10s	Quorum stability window before returning to normal
Stake visibility threshold	66.66%	Minimum visible stake for normal operation

The **PartitionDetector** runs every 5 seconds, computing the percentage of total stake reachable via healthy gossip peers. Two detection signals are used:

1. **Visible stake percentage.** The primary signal. The detector queries the gossip service for currently connected peers, maps them to their validator identities, and sums their stake. If the sum falls below 2/3 of total stake, the detector transitions to SUSPECTED.
2. **Checkpoint stall detection.** A secondary signal. If no new checkpoint has been finalized for 3x the expected checkpoint interval and the node has unfinalized transactions, this indicates a likely quorum failure. This signal catches cases where visible stake calculations are stale due to peer-list update delays.

The SUSPECTED state serves as a damping buffer - transient network hiccups (brief disconnections, route changes) do not trigger partition mode unless they persist for T_conf seconds. This prevents unnecessary mode switching in mildly unstable networks.

8.2 Partition Epochs

Upon entering PARTITIONED mode, a monotonically increasing `partition_epoch` is assigned. This epoch tags all transactions and provisional checkpoints created during the partition, enabling the merge protocol to distinguish pre-partition from during-partition state.

8.3 Provisional Checkpoints

During a partition, the checkpoint quorum threshold relaxes from 2/3 of total stake to **2/3 of visible stake** (minimum 1/3 of total stake as a safety floor). Checkpoints created under this relaxed quorum carry:

- `provisional: true`
- `partition_epoch: <epoch>`
- `visible_stake_pct: <fraction>`

Transactions finalized by provisional checkpoints receive **provisional finality** (see Section 2.3). They are treated as finalized for local operations but remain rollback-eligible during merge. Clients and applications are explicitly informed of the provisional status.

The safety floor of 1/3 total stake prevents a single isolated node from creating provisional checkpoints unilaterally - there must be a meaningful fraction of the validator set present in the partition for provisional operation to proceed.

8.4 Local Operation During Partition

All core operations continue during a partition, subject to the transaction classification system (Section 9.8):

- **Safe transactions** (DataOnly, Consolidation, Reward): Proceed without restriction. These are inherently merge-safe.

- **BoundedSpend transactions** (Transfer, Contract): Proceed up to the account's partition budget limit, if one is configured. Without a budget, they proceed without restriction but carry rollback risk.
- **CpOnly transactions** (Stake, Unstake, ClaimRewards): Rejected during partition. These operations modify the validator set or consensus-critical state and require full quorum for safety.

The partition is transparent to the application layer except for finality semantics (provisional vs. final) and CpOnly rejection. DAG tip selection, weight calculation, gas pricing, and smart contract execution operate identically in both modes.

8.5 Provisional Receipt Semantics

During partition mode, all receipts and VerifiableObjects carry the `partition_epoch` and `provisional: true` markers. A `ProofFreshness` attached to any VO generated during partition mode includes the partition epoch, enabling downstream consumers to make informed trust decisions. Applications that require settlement finality can choose to wait for post-merge reconciliation; applications that prioritize responsiveness (messaging, social features) can accept provisional receipts immediately.

9. Reconciliation Semantics & Transaction Taxonomy

This section describes the core protocol innovation: a deterministic 5-phase merge protocol that reconciles divergent partition state while preserving maximum valid work from all partitions. This section is intended to be the most rigorous in the paper and will be the primary target of formal analysis.

Scope assumption: two-partition merge. The current merge protocol is designed and analyzed for the two-partition case: a single network split that produces two independent partition groups, which later reconnect and reconcile. Three-way or multi-way partitions (where the network fragments into three or more independent groups simultaneously) require a fundamentally different merge coordination protocol — pairwise merge of three independent provisional chains may not commute, and the merge order could affect the final state. For the current protocol version, multi-way partitions are handled by sequential pairwise merges (each reconnecting pair merges independently, with subsequent merges treating the merged state as the new baseline). Formal analysis of commutativity guarantees for sequential pairwise merge, and a native multi-party merge protocol, are future work.

9.1 Determinism Requirements

All merge computation must produce identical results on every node given the same inputs. This is the foundational constraint - without it, nodes would diverge after merge, which is worse than the partition itself.

Integer accounting: All balance operations during merge use **u64** micro-units (1 RKU = 10^8 micro-RKU). Balances are converted from **f64** to micro-units at merge entry and converted back only after reconciliation is complete. This eliminates floating-point nondeterminism entirely within the merge path.

Canonical ordering: Transactions are totally ordered by:

1. Per-account nonce ascending
2. DAG depth from fork-point (topological distance)
3. Transaction hash lexicographic ascending (deterministic tiebreaker)

This is a strict total order - no two transactions can be "equal" because transaction hashes are unique. Every node computing this order on the same transaction set produces the identical sequence.

Contract isolation: Contract calls are **not re-executed** during merge. The merge algorithm operates only on the balance transfers recorded in the original transaction execution (amount + gas fee). This avoids nondeterminism from contract execution order, memory state, or environmental dependencies.

Contract storage conflicts are resolved separately via "last-write-wins by weight" on individual storage keys. For each key written by transactions in both partitions, the write from the transaction with higher cumulative weight wins.

Known limitation: atomic multi-key updates. The per-key "last-write-wins" rule may produce inconsistent contract state when contract logic depends on multiple storage keys being written atomically. For example, if a contract maintains an invariant **balance_a + balance_b == total** and Partition A updates **balance_a** while Partition B updates **balance_b**, the merged state may violate the invariant. Mitigation strategies include contract-level merge hooks (Section 15.4), storage namespacing by partition safety class, and restricting AP-mode contract calls to safe subsets via the transaction classification system.

9.2 Phase 1 - DAG Exchange

Both partitions exchange their **PartitionDAGDelta** containing all transactions, provisional checkpoints, and DAG edges created since the fork point (last common confirmed checkpoint). After exchange, both sides have the complete merged transaction set.

Exchange is performed via gossip protocol messages (`GossipMessage::MergePayload` and `GossipMessage::MergeResult`). The merge payload includes a `MergeRequest` containing:

- All transactions created during the partition epoch
- Account state snapshots from the partition
- Provisional checkpoint chain
- Fork point reference (last common confirmed checkpoint height and hash)

9.3 Phase 2 - Conflict Detection

The merged transaction set is scanned for two conflict types:

Type 1 - Direct Double-Spend: Same account + same nonce in both partitions. This requires the user to have deliberately crafted a conflicting transaction - it cannot happen accidentally because nonces are sequential. Unambiguous evidence of intentional double-spending.

Type 2 - Economic Overdraft: No nonce collision, but combined spending from both partitions exceeds the account's pre-partition balance. Example: Alice had 100 RKU at the fork point, spent 60 RKU (nonce 5) in Partition A and 60 RKU (nonce 5 in Partition B, which was locally valid because Partition B never saw nonce 5 from Partition A). Combined: 120 spent on 100 balance.

This may be innocent (the user didn't know about the partition and transacted on both sides) or opportunistic.

Detection algorithm:

1. Build a set of all (`account`, `nonce`) pairs across both partitions. Flag any duplicates as Type 1.
2. For each account that transacted in both partitions, compute pre-partition balance (from fork-point checkpoint), total sent and received in each partition. If combined net spend exceeds pre-partition balance, flag as Type 2.

9.4 Phase 3 - Weight Resolution

For each conflict, a deterministic winner is selected:

Direct double-spends: Winner is the transaction with higher cumulative DAG weight in the merged graph. If weights are within the `WEIGHT_PROXIMITY` threshold (1.5x), a tiebreaker applies. Formally, two weights `W_a` and `W_b` are considered "proximate" when $\max(W_a, W_b) / \min(W_a, W_b) < 1.5$ — i.e., neither weight exceeds 1.5 times the other. When weights are proximate, the tiebreaker is the partition with higher `visible_stake_pct` at the

time of the provisional checkpoint that finalized the transaction. If still tied, the transaction with the lexicographically lower hash wins.

Economic overdrafts: All transactions from the conflicting account are ordered by nonce ascending, then cumulative weight descending. Starting from the pre-partition balance (in micro-units), transactions are replayed in this order. The first transaction that would cause an overdraft - and all subsequent transactions from that account - are marked as losers.

9.5 Phase 4 - Cascade Rollback

The key insight: rejecting a transaction doesn't just affect its sender - it affects everyone who received funds from that transaction and subsequently spent them. The cascade rollback algorithm traces these economic dependency chains.

Algorithm (pseudocode):

```
function cascade_rollback(losing_txs, all_txs):

    balances = snapshot_balances_at_fork_point()    // u64 micro-units

    surviving_txs = all_txs.exclude(losing_txs)

    ordered_txs = canonical_sort(surviving_txs)    // see Section 9.1

    rolled_back = Set()

    loop:

        newly_rolled_back = Set()

        balances = snapshot_balances_at_fork_point() // reset each iteration

        expected_nonce = {}                        // per-account

        for tx in ordered_txs:

            if tx.hash in rolled_back: continue

            // Nonce continuity: if a prior nonce was rolled back,

            // all subsequent nonces from this account are invalid

            if tx.nonce > expected_nonce[tx.from]:
```

```

        newly_rolled_back.add(tx.hash)

        continue

    cost = to_micro(tx.amount) + to_micro(tx.gas)

    if balances[tx.from] < cost:

        newly_rolled_back.add(tx.hash)

        continue

    balances[tx.from] -= cost

    balances[tx.to] += to_micro(tx.amount)

    expected_nonce[tx.from] += 1

    if newly_rolled_back.is_empty(): break    // stable

    rolled_back.extend(newly_rolled_back)

return (rolled_back, balances)

```

Convergence guarantee: Each iteration can only add rollbacks, never remove them. The valid transaction set shrinks monotonically. Since the transaction set is finite, the algorithm terminates.

Proof of convergence (termination). Let S_i be the set of surviving transactions after iteration i . The algorithm guarantees $S_{i+1} \subseteq S_i$ (monotonic shrinkage): each iteration either produces at least one newly rolled-back transaction ($|S_{i+1}| < |S_i|$) or produces no new rollbacks ($S_{i+1} = S_i$, and the algorithm terminates). Since $|S_i| \geq 0$ and decreases by at least 1 per non-terminal iteration, the algorithm terminates in at most $|S_0|$ steps.

Proof of convergence (correctness). Every node computing the cascade on the same input set produces identical output because: (1) the initial losing set is determined by the deterministic conflict resolution algorithm (Section 9.4); (2) `snapshot_balances_at_fork_point()` returns the same u64 micro-unit balances on all nodes (anchored to a common confirmed checkpoint); (3) `canonical_sort` produces an identical total order on all nodes (Section 9.1); (4) the replay loop is a pure deterministic function — for each transaction in canonical order, the balance check (`balances[tx.from] < cost`) and nonce continuity check (`tx.nonce > expected_nonce[tx.from]`) produce identical boolean results given identical input state; (5)

because balances are reset from the fork-point snapshot at the start of each iteration and the rolled-back set only grows, the inputs to each iteration are fully determined by the growing `rolled_back` set, which is itself deterministic. Therefore S_i is identical on all nodes for all i , and the final state (balances and surviving transaction set) is identical.

Complexity: Worst case $O(n^2)$ where n is the number of transactions in the partition period. Each iteration is $O(n)$ (replay all transactions), and at most n iterations occur. In practice, cascades are shallow — empirically, most transactions do not depend on funds from rolled-back transactions, and convergence occurs in 1–3 iterations. For a partition with k direct conflicts and an average dependency chain depth of d , the expected iteration count is $O(d)$, which is typically $O(1)$ for well-distributed transaction graphs.

9.6 Phase 5 - State Reconciliation

After cascade rollback completes:

1. **Account state rebuild:** `final_balances` from the cascade replay (in micro-units) are converted back to `f64` and written as canonical account state.
2. **DAG cleanup:** Direct conflict losers (nonce reuse - malicious) are removed from the DAG entirely. Cascade victims (innocent users whose upstream funds evaporated) are marked `rolled_back: true` and kept for auditability.
3. **Merge checkpoint:** A new checkpoint is created at `fork_point_height + 1` with:
 - `provisional: false`
 - `previous_hash` linking to the fork-point checkpoint
 - `finalized_tx_hashes` containing all surviving transactions from both partitions
 - `merge_report_hash` referencing the full MergeReport
 - Signed by the reunified validator quorum (must meet full 2/3 total stake threshold)
4. **Provisional chain retirement:** All provisional checkpoints from both partitions are archived to `merge_history` and removed from the active checkpoint chain.

9.7 Nonce Gap Behavior

When a transaction is rolled back, all subsequent transactions from that account (with higher nonces) are also rolled back, regardless of their individual validity. This is because Rinku enforces strict nonce sequentiality - a "gap" in the nonce sequence means the account's state is undefined for all operations after the gap.

This is a deliberate design choice: it prevents subtle state inconsistencies where an account's later transactions depend on side effects (balance changes, contract state mutations) of the rolled-back transaction, even if the later transactions appear independently valid.

Why nonce-gap rollback is necessary. Consider the alternative - nonce remapping, where surviving transactions are renumbered to fill gaps. This introduces several problems: (1)

transaction hashes would change (the nonce is part of the signed data), invalidating signatures; (2) any VerifiableObject referencing the original transaction hash would become invalid; (3) the remapped transaction would need re-execution for contracts, violating the determinism requirement. Partial replay (skipping the gap and continuing with later nonces) is equally problematic because later transactions may depend on state changes from the skipped transaction - a transfer at nonce 3 may only be valid because nonce 2 received incoming funds.

Nonce-gap rollback is conservative - it potentially rolls back more transactions than strictly necessary. This is an intentional safety-over-liveness tradeoff: the cost of unnecessary rollbacks (user inconvenience) is far lower than the cost of state corruption from invalid partial replays.

Important subtlety: cross-partition nonce filling. When a transaction loses a direct conflict (e.g., local nonce 1 loses to remote nonce 1), the winning transaction fills the nonce slot. Subsequent nonces (2, 3, ...) from the same account are NOT rolled back - the sequence remains unbroken. Nonce-gap cascades only occur when no transaction (from either partition) fills a nonce slot.

9.8 Transaction Classification

Rinku implements a **PartitionSafety** classification system that governs which transactions may execute during partition mode. This is a protocol-level enforcement, not an application-layer convention - the gate is applied in the transaction acceptance path before DAG insertion.

Classification	AP-Mode Behavior	Transaction Types	Rationale
Safe	Always allowed	DataOnly, Consolidation, Reward	Append-only or system-generated; no cross-account economic state mutation; inherently merge-safe
BoundedSpend	Allowed within partition budget	Transfer, Contract	May create cross-partition conflicts; bounded by optional per-account spending limit
CpOnly	Rejected during partition	Stake, Unstake, ClaimRewards	Modify validator set or consensus-critical state; require full quorum for safety

Partition budget system. Accounts can optionally configure a `partition_budget` - a maximum amount spendable during any single partition epoch. When a BoundedSpend transaction is submitted during partition mode, the protocol checks:

1. Has the account configured a partition budget? If not, the transaction proceeds without restriction (but carries full rollback risk).
2. Would this transaction cause `partition_budget_spent + tx_amount` to exceed `partition_budget`? If so, the transaction is rejected.

Transactions within the partition budget are economically guaranteed to be merge-safe: even in the worst case (identical spending in both partitions), the combined spend cannot exceed the pre-partition balance if the budget is set to at most half the balance. This transforms partition tolerance from a probabilistic property to a deterministic one for opted-in accounts.

The budget is tracked via `partition_budget_spent` on the Account struct and is reset when the node enters a new partition epoch.

9.9 Three-Tier Receipt Model

The VerifiableObject system supports explicit receipt tiers as semantic markers:

- **TentativeReceipt:** Issued during partition mode. Carries provisional checkpoint anchor and partition epoch. Explicitly communicates "valid locally, subject to rollback." Applications can render these with appropriate UI affordances (e.g., a pending indicator).
- **FinalReceipt:** Issued after checkpoint finality during normal operation. Carries full BLS quorum signature. Irrevocable under honest majority assumption.
- **ReconciliationReceipt:** Issued after merge. Proves that a tentative transaction was either accepted (upgraded to final status in the merge checkpoint) or rejected (rolled back with reason code: `ConflictLoser`, `CascadeVictim`, `NonceContinuityGap`, `InsufficientBalanceAfterConflictResolution`).

These tiers are not separate data structures but semantic interpretations of the existing `VerifiableObject` with its `provisional`, `partition_epoch`, and `merge_report_hash` fields. Applications and wallets should inspect these fields to determine the receipt tier and display appropriate finality information to users.

10. Smart Contracts & WASM Runtime

10.1 Runtime Architecture

Rinku executes smart contracts in a sandboxed WASM environment built on the `wasmi` interpreter:

- **Memory sandbox:** Configurable up to 256 pages (16 MB), with bounds-checked guest memory access
- **Import validation:** Only `rinku` and `env` namespaces are permitted; all other imports are rejected at deployment
- **Deterministic execution:** `wasmi` provides bit-identical execution across platforms

10.2 Dual Gas Metering

Gas is metered at two levels to prevent both instruction-level abuse and host-call abuse:

- **Fuel metering:** Every WASM instruction consumes interpreter fuel (default budget: 10,000,000 fuel units)
- **Host gas metering:** Expensive host operations charge additional gas through the `GasMeter`

Total gas = $(\text{fuel_consumed} / 100) + \text{host_gas} + \text{base_gas} + \text{input_bytes_gas}$

Where $\text{input_bytes_gas} = \text{input_size} * 16$ (16 gas per byte of contract input).

Host operation gas schedule:

Operation	Gas Cost	Rationale
<code>base_execution</code>	1,000	Fixed cost per contract invocation
<code>storage_read</code>	200	Database lookup
<code>storage_write</code>	5,000	Database write + state hash update
<code>storage_delete</code>	5,000	Database delete + state hash update
<code>memory_alloc</code>	3	Per-page memory allocation

Operation	Gas Cost	Rationale
<code>transfer</code>	8,000	Balance mutation on two accounts
<code>mint</code>	6,000	Token minting operation
<code>burn</code>	6,000	Token burning operation
<code>emit</code>	500	Event serialization and broadcast
<code>hash</code>	300	Cryptographic hash (sha256/keccak256)
<code>balance_check</code>	100	Ledger state lookup (get_balance/get_staked)
<code>log</code>	100	Debug output (no state mutation)

10.3 Host ABI

The `rinku` namespace exposes:

Function	Description
<code>storage_read, storage_write, storage_delete, storage_has</code>	Contract key-value storage (JSON-serialized values)
<code>get_caller, get_block_height, get_timestamp, get_input</code>	Execution context
<code>get_contract_id</code>	Self-referential contract address
<code>get_balance, get_staked</code>	Ledger queries
<code>transfer</code>	Native token transfers from contract balance
<code>emit_event</code>	Event emission for indexing and WebSocket subscribers
<code>emit_view_key</code>	Expose state fragments for stateless verification
<code>sha256, keccak256</code>	Cryptographic hashing

10.4 Stateless dApp Standard (Proof-Carrying Contracts)

Contracts define `ViewKeySpec` schemas specifying which pieces of state should be exposed for external verification. Every mutating call returns a `StatefulReceipt` containing:

- View key values (specific state fragments selected by the contract)
- Merkle multiproof connecting those values to the checkpoint state root
- Finality certificate (checkpoint anchor with BLS signature)

This enables **persistently stateless clients** - applications that never store blockchain state locally but can verify any claim on demand using receipts. Concretely, a persistently stateless client works as follows: when the application needs to display data (a user's balance, a social feed, a contract's leaderboard), it queries any Rinku node for the relevant `StateWitness` - a sparse Merkle multiproof over the specific storage keys it needs, anchored to the latest finalized checkpoint. The node returns the witness as a VerifiableObject URL. The client verifies the Merkle proof against the checkpoint's state root and the BLS aggregate signature against the known validator set, then renders the proven data directly. It stores nothing. The next time it needs the same data, it requests a fresh witness. There is no local database, no sync process, no stale cache to invalidate. The client is stateless not just at startup but persistently - it never accumulates state that could drift from the canonical chain. The tradeoff is that the client must be able to reach at least one node to obtain fresh witnesses; it cannot serve reads while fully offline (though it can verify previously-received VOs offline indefinitely).

10.5 Contract SDK

Rinku provides a Rust SDK (`rinku-contract-sdk`) for contract development with the following macros and helpers:

- **`entrypoint!`** - Declares the contract's entry point function, handling input deserialization and output serialization.
- **`contract_init!`** - Declares the contract's initialization function, called once at deployment.
- **`contract_call!`** - Declares a callable contract function with automatic gas metering and error handling.

The SDK provides helper functions for common operations:

- **Storage:** `storage::get<T>()`, `storage::set<T>()`, `storage::delete()`, `storage::has()` - type-safe wrappers around the host ABI storage functions with automatic JSON serialization.
- **Transfers:** `token::transfer(to, amount)` - transfers RKU from the contract's balance.

- **Events:** `events::emit(name, data)` - emits a named event for indexing and WebSocket subscribers.
- **View keys:** `view::expose(key, value)` - registers a view key for inclusion in the `StatefulReceipt`.

Contracts are compiled to WASM using standard Rust toolchain targeting `wasm32-unknown-unknown`, then deployed via a `Contract` transaction with the base64-encoded WASM binary and initial state.

11. Economic Model

11.1 Supply Schedule

Parameter	Value
Maximum supply	30,000,000 RKU
Genesis allocation	6,000,000 RKU (20%)
Maximum emittable	24,000,000 RKU
Halving interval	3,150,000 checkpoints (~1 year at 10s intervals)
Total halvings	5
Minimum reward floor	0.122887 RKU per checkpoint (= $\text{initial_reward} / 2^5$, permanent floor that does not halve further)

Genesis allocation distribution. The 6,000,000 RKU genesis allocation is distributed as follows: each of the 3 genesis validators receives an initial balance sufficient to stake `GENESIS_VALIDATOR_STAKE` (50,000 RKU each, totaling 150,000 RKU staked). The remaining 5,850,000 RKU constitutes the genesis reserve, held in the genesis account for future distribution via governance (not yet implemented), ecosystem grants, and faucet funding on testnet. No single entity holds a controlling share of the genesis allocation — the maximum individual holding at genesis (if all reserve were concentrated) would be ~19.5% of total supply, insufficient for unilateral quorum influence (requires >33.33%).

Halving interval derivation. The interval of 3,150,000 checkpoints is derived from the target of approximately 1 year per halving epoch at the design checkpoint interval of 10 seconds:

$365.25 \text{ days} \times 24 \text{ hours} \times 3600 \text{ seconds} / 10 \text{ seconds} \approx 3,155,760$, rounded to 3,150,000 for a clean constant. Five halvings produce a geometric emission decay (initial reward $\sim 3.93 \text{ RKU} \rightarrow 1.965 \rightarrow 0.983 \rightarrow 0.491 \rightarrow 0.246$), after which the reward floor of 0.122887 RKU applies permanently — this floor does not halve further, ensuring perpetual validator incentives. With the floor in place, total supply asymptotically approaches but never exceeds 30,000,000 RKU because the emission logic enforces $\min(\text{reward}, \text{MAX_SUPPLY} - \text{current_supply})$.

Emission schedule:

Halving	Checkpoints	Reward per Checkpoint	Cumulative Emited
0	0 – 3,149,999	$\sim 3.93 \text{ RKU}$	$\sim 12,379,500 \text{ RKU}$
1	3,150,000 – 6,299,999	$\sim 1.965 \text{ RKU}$	$\sim 18,569,250 \text{ RKU}$
2	6,300,000 – 9,449,999	$\sim 0.983 \text{ RKU}$	$\sim 21,664,125 \text{ RKU}$
3	9,450,000 – 12,599,999	$\sim 0.491 \text{ RKU}$	$\sim 23,211,563 \text{ RKU}$
4	12,600,000 – 15,749,999	$\sim 0.246 \text{ RKU}$	$\sim 23,985,281 \text{ RKU}$
5+	15,750,000+	0.122887 RKU (floor)	Approaches 30,000,000

The emission curve follows a geometric decay with a floor, ensuring that block rewards never reach zero - validators always have a base-layer incentive to participate, even after the majority of tokens have been emitted. The hard cap of 30M RKU is asymptotically approached but never exceeded due to the floor mechanism and total supply enforcement in the emission logic.

11.2 Checkpoint Rewards

Checkpoint rewards are distributed when a checkpoint is finalized. The reward amount is determined by the emission schedule (Section 11.1) and is distributed to all active validators proportional to their effective weight:

Distribution formula:

$$\text{validator_reward} = \text{checkpoint_reward} * (\text{validator_effective_weight} / \text{total_effective_weight})$$

Where `effective_weight` is computed using the dual-weight system (Section 11.3). The total checkpoint reward is minted as new supply, subject to the hard cap - if minting the full reward would exceed 30,000,000 RKU, the reward is reduced to the remaining mintable amount.

Rewards are credited directly to the validator's liquid balance (not to their staked balance), allowing validators to compound their stake through explicit re-staking or use rewards for other purposes.

11.3 Weighted Proof-of-Stake (WPoS) Reward Distribution

Checkpoint rewards are distributed through a dual-weight system. Note: the "weight" in WPoS reward distribution is distinct from the DAG transaction weight described in Section 7.3. DAG weight determines transaction ordering priority and conflict resolution; WPoS weight determines validator reward shares.

- **Stake weight (70%):** Proportional to amount staked. The 70/30 split is calibrated to ensure that stake remains the dominant factor in reward distribution (preventing age-only validators from receiving outsized rewards) while providing a meaningful incentive for continuous participation.
- **Age weight (30%):** Rewards long-term active participation; requires minimum 100 RKU bond; decays 10% per missed checkpoint. Age weight is computed as $\min(\text{checkpoints_since_stake} / \text{TARGET_AGE}, 1.0)$ where `TARGET_AGE` is 1000 checkpoints (~2.7 hours). This ramps linearly from 0 to 1.0 over the target period, rewarding validators who maintain continuous uptime. The 10% decay per missed checkpoint penalizes intermittent validators.

11.4 Staking Requirements

Parameter	Value
Minimum stake	100 RKU
Minimum stake age for rewards	15 seconds
Unstake cooldown	24 hours
Slashing unbonding period	14 days

11.5 Additional Reward Streams

- **Tip rewards (1%):** Distributed as incentives for specific network actions
- **Witness rewards (0.2%):** Incentivize DAG connectivity by rewarding nodes that reference other transactions as parents

11.6 Gas Fee Model

Rinku implements an EIP-1559-inspired dynamic gas pricing mechanism:

Parameter	Value
Target throughput	150 transactions per 15-second period
Adjustment factor	12.5% per period
Minimum gas price	0.001 RKU
Maximum gas price	10.0 RKU

The gas price adjusts each period: if the actual transaction count exceeds the target, the price increases by the adjustment factor; if below target, it decreases. This creates a self-regulating fee market that responds to demand without requiring explicit fee auctions.

11.7 Adaptive Fee Burn

Transaction fees are split between burning and validator rewards:

- **Burn ceiling:** 30% of fees
- **Burn scaling:** Burn percentage increases linearly as circulating supply approaches 50% of the hard cap. At 0% of max supply, no fees are burned; at 50% (15M RKU), the full 30% ceiling is reached.
- **Validator floor:** Validators always receive at least 70% of fees

The adaptive burn creates deflationary pressure that increases as the token supply grows, counterbalancing emission inflation and creating a natural equilibrium. At maturity (when emission is at the floor rate), the burn mechanism may offset or exceed emission, potentially making the token supply effectively stable or mildly deflationary depending on transaction volume.

11.8 Micro-Unit Precision

All internal accounting uses **u64** micro-units with 8 decimal places (1 RKU = 100,000,000 micro-RKU). This eliminates floating-point precision errors, which is particularly critical for deterministic merge reconciliation (Section 9).

Boundary disclosure. The **AccountStateProof** API response struct intentionally carries both canonical **u64** micro-unit fields (**balance_micro**, **staked_micro**) and **f64** display fields (**balance**, **staked**) for JSON readability. The **f64** fields are derived from micro-units via

`from_micro_units()` at serialization time and are provided for human consumption only. All consensus-critical operations — merge reconciliation, balance checks, gas enforcement, reward calculation — operate exclusively on `u64` micro-units. Cross-network state exchange (sync snapshots, gossip messages) also uses `u64`. The `f64` representation is lossy for values exceeding 2^{53} micro-units (~90,071,992 RKU), which is above the hard cap and therefore does not affect correctness in practice.

12. Slashing & Economic Security

12.1 Violation Types

Violation	Severity	Penalty
Nonce Reuse (Cross-Partition Double-Spend)	Malicious	10% balance penalty + 100% stake slash + 0.50 permanent reputation penalty
Cross-Partition Economic Overdraft	Gray area	0.10 reputation penalty with linear decay over 100 checkpoints
Cascade Victim	Innocent	No penalty
Double-Sign (same height, different hash)	Malicious	15% stake slash
Invalid Checkpoint Proposal	Malicious	25% stake slash
Receipt/Proof Tampering	Malicious	25% stake slash
Invalid Proof/Witness Submission	Malicious	15-20% stake slash
Liveness Failure (3+ missed checkpoints)	Negligent	5% stake slash (increasing to 10% for repeat offenses)

The graduated penalty structure reflects a key design principle: **the protocol distinguishes intent**. Nonce reuse requires deliberate action and is treated as malicious. Economic overdraft may be accidental and receives a soft, recoverable penalty. Cascade victims are blameless and bear no cost.

12.2 Reputation & Weight Modifier

Accounts with reputation penalties receive reduced weight in all weight calculations:

$$\text{effective_weight} = \text{base_weight} * (1.0 - \text{reputation_penalty})$$

This reduces the influence of penalized accounts in DAG weight calculations, tip selection, and consensus voting without requiring immediate ejection from the network.

12.3 Penalty Decay

Cross-partition overdraft reputation penalties decay linearly over 100 checkpoints (~16 minutes at 10s intervals), allowing honest users caught in an ambiguous situation to recover their standing. Nonce-reuse penalties are permanent and do not decay.

The decay is computed as:

$$\text{remaining_penalty} = \text{original_penalty} * \max(0, 1.0 - (\text{current_height} - \text{penalty_height}) / 100)$$

Where `penalty_height` is the checkpoint at which the penalty was applied (stored in `penalty_decay_checkpoint` on the Account struct).

12.4 Game-Theoretic Analysis

Nonce-reuse attack cost. An attacker attempting a cross-partition double-spend faces: 10% balance confiscation + 100% stake slash + 0.50 permanent reputation penalty. For this to be profitable, the double-spent amount must exceed $0.10 * \text{balance} + 1.00 * \text{stake} + \text{NPV}(\text{weight_reduction})$. The permanent reputation penalty means the attacker's future transaction weight is halved indefinitely, reducing their influence in all protocol interactions. For any account with meaningful stake, the expected cost far exceeds the maximum double-spend gain (which is bounded by the account's pre-partition balance).

Economic overdraft opportunism. An attacker who intentionally spends in both partitions faces a 0.10 reputation penalty with 100-checkpoint decay. The cost is temporary weight reduction. This penalty is intentionally mild because the attack surface is limited (the attacker can only spend their own balance) and the behavior may be innocent. The partition budget system provides a stronger mitigation for users who want guaranteed merge safety.

Cascade attack analysis. An attacker cannot deliberately cause cascades against specific victims without first losing their own funds in a direct conflict. Cascade rollbacks are a second-order effect - they require the attacker to sacrifice their own transaction first. The attacker bears the full penalty for the initiating conflict; cascade victims bear no penalty. The

attacker cannot profit from cascading because the rolled-back funds return to their pre-partition state, not to the attacker.

12.5 Acknowledged Security Considerations

The following security properties are areas of active analysis and represent known open questions in the current protocol design:

Nothing-at-stake for convergence ACKs. Validators have no explicit penalty for selective ACK withholding — a validator can delay or omit ACKs for competitor transactions to slow their convergence confirmation. Mitigation: convergence is the primary finality mechanism, so withheld ACKs can delay transaction finality. However, liveness failure penalties (Section 12.1) provide indirect deterrence for persistent non-participation, and the 2/3 quorum threshold means any individual validator's withholding is insufficient to prevent convergence as long as the remaining stake exceeds the threshold. Formal analysis of selective withholding incentives is future work.

Partition budget replay via rapid cycling. The partition budget resets per partition epoch. An adversary who can rapidly induce partition cycling (NORMAL → PARTITIONED → NORMAL → PARTITIONED) could multiply the effective budget by the number of cycles. Mitigation: the T_conf (30s) and T_recovery (10s) timeouts impose a minimum 40-second cycle time, bounding the cycling rate. Additionally, rapid cycling itself requires control over network connectivity affecting >1/3 of stake, which is a strong adversary assumption. A minimum inter-epoch delay or cumulative budget across epochs is under consideration.

Merge strategic delay. A validator whose transactions will lose conflict resolution has an incentive to delay broadcasting their MergePayload. Currently, no explicit timeout enforcement or penalty for delayed merge participation is described. In practice, merge is orchestrated by the reconnecting nodes and does not require the cooperation of the validator whose transactions lose — the winning partition's state is adopted. Formal timeout-based merge liveness guarantees are future work.

Slashing evidence authentication. `SlashingEvidence` gossip messages carry the conflicting signed messages as proof. The evidence itself is self-authenticating — the conflicting signatures are verifiable by any node against the validator's public key. However, the gossip message wrapper is not signed by the evidence submitter, meaning any node can relay (but not forge) slashing evidence. This is by design: evidence should be freely relayable, and forgery is impossible because the evidence contains the validator's own signatures.

VO proof replay within freshness window. The `max_age_checkpoints` mechanism prevents stale proof replay but does not prevent repeated submission of the same valid proof within the freshness window. For BYOP contract interactions, this means a contract must implement its own replay protection (e.g., tracking consumed proof hashes) if single-use

semantics are required. The protocol provides the freshness primitive; application-layer replay protection is the contract's responsibility.

BYOP front-running. The receipt-composition pattern (Section 3.4) is vulnerable to front-running: an attacker observing a pending BYOP transaction can submit a competing transaction with a more favorable oracle receipt within the freshness window. This is a general MEV concern shared with all receipt-composable systems. Mitigation strategies include shorter freshness windows, commit-reveal schemes at the contract layer, and encrypted mempool proposals (future work).

Signer bitmap authentication. The signer bitmap identifying which validators signed a checkpoint is not independently signed — it is embedded within the checkpoint data structure alongside the aggregate BLS signature. The bitmap's integrity is implicitly verified: if the bitmap is tampered with (claiming additional signers), the aggregate public key reconstruction will not match the aggregate signature, and verification will fail. An attacker cannot inflate the apparent quorum without possessing the corresponding BLS private keys.

Validator set evolution for offline VO verification. A VO generated at checkpoint N cannot be verified offline by a party holding only the genesis validator set if the validator set changed between genesis and checkpoint N. Offline verification requires the verifier to hold a validator set that was active at the VO's checkpoint height. For long-lived offline verification, VOs should embed the signer membership proofs (Section 4.2, item 5) that chain back to a known anchor. Full validator set evolution proofs (chaining validator set changes through signed checkpoints) are future work.

Validator exit before slashing. A validator that double-signs could attempt to front-run slashing by submitting an unstake transaction during the 24-hour cooldown. Mitigation: the 14-day unbonding period (Section 11.4) means staked tokens remain slashable for 14 days after unstaking, regardless of when the unstake was initiated. Slashing evidence submitted during the unbonding period is applied to the locked stake. A double-signer cannot escape slashing by unstaking.

Sub-linear stake and Sybil resistance. The sub-linear stake weight formula ($\text{stake}^{0.5} * 2.0$) means that splitting stake across N identities yields higher aggregate weight than a single large stake (N validators at stake S/N each have total weight $N * (S/N)^{0.5} * 2.0 = 2.0 * S^{0.5} * N^{0.5}$). Sybil resistance is therefore load-bearing on the minimum stake requirement (100 RKU): creating N Sybil validators requires $N * 100$ RKU, limiting the splitting advantage. The practical break-even point where splitting becomes profitable depends on the attacker's total stake relative to the minimum.

13. Networking & P2P Protocol

13.1 Gossipsub

Rinku's gossip protocol operates on the `rinku/1.0.0` topic with the following message types:

Message Type	Priority	Description
Transaction	Normal	Signed transaction propagation
TipAnnouncement	Normal	Current DAG tips and size (triggers sync if peer is ahead)
CheckpointAnnouncement	High	New finalized checkpoint with transaction bodies
CheckpointSignature	High	Individual validator signature for checkpoint voting
TxConfirmBroadcast	High	Convergence broadcast for per-transaction finality
TxConfirmAck	High	Validator acknowledgment for convergence voting
BloomAnnouncement	Low	Bandwidth-efficient advertisement of known transactions
PeerDiscovery	Low	Shares known peer addresses for mesh expansion
ConflictResolution	Normal	Broadcasts conflict resolution decisions
SlashingEvidence	High	Proof of validator misbehavior (double-signing)
WeightVote	Normal	Validator vote for transaction trust weighting

Message Type	Priority	Description
MergePayload	High	Partition merge request with DAG delta
MergeResult	High	Partition merge response with reconciliation data
SyncRequest	Normal	Request for missing transactions
SyncResponse	Normal	Response with requested transactions

Bloom filter optimization. To reduce bandwidth, nodes periodically broadcast `BloomAnnouncement` messages containing a Bloom filter of their known transaction hashes. The filter uses double SHA-256 hashing for optimal bit distribution. Before sending a transaction to a peer, the node checks the peer's Bloom filter - if the transaction is likely already known, it is not retransmitted. The filter includes the node's checkpoint height and tip count, enabling peers to detect if they are behind and should initiate sync.

Propagation batching. To prevent message storms under high transaction load, the gossip service uses a background propagation task with a `MAX_PROPAGATION_BATCH` of 100 transactions. Pending transactions are accumulated and broadcast in batches, amortizing the per-message overhead of gossipsub.

Deduplication. A `BoundedHashSet` (`known_txs`) tracks recently seen transaction hashes to prevent infinite gossip loops. The set has a bounded capacity and evicts oldest entries when full.

13.2 Lock-Free Message Handling

Rinku's P2P receive path is designed to eliminate mutex contention on the critical message-processing path:

Channel-based architecture. The `NetworkHandle` wraps the libp2p swarm and exposes message channels. During initialization, the gossip message receiver (`message_rx`) and sync request receiver (`sync_incoming_rx`) are extracted from the `NetworkHandle` **before** it is wrapped in `Arc<Mutex<>>`. These channels are passed directly to handler tasks via `set_p2p_channels()`.

Implications. The message receive path never acquires the `NetworkHandle` mutex. Incoming gossip messages flow from the libp2p event loop → mpvc channel → `run_p2p_receiver` task

→ `handle_message` processing, entirely lock-free. Similarly, sync requests flow through a separate channel to `run_sync_request_handler`. Sync response sending uses a cloned `command_tx` sender, which is also lock-free (mpsc senders are cheaply cloneable).

This architecture eliminates the 5-25ms polling latency per message that would occur if the receive path needed to acquire a mutex on every incoming message, which is critical for maintaining sub-second convergence finality latency under load.

13.3 Connection Management

Idle timeout. Connections have a 600-second idle timeout, configured to prevent premature `KeepAliveTimeout` disconnects on low-traffic deployments where minutes may pass between gossip messages.

Mesh maintenance. The network service periodically checks if the number of validated peers is below `MIN_MESH_PEERS`. If the mesh is unhealthy, it re-dials bootstrap peers to restore connectivity. The `InsufficientPeers` publish error (which occurs during startup or reconnection when no gossipsub peers are available) is logged at trace level to avoid log noise during expected transient states. Note: `MIN_MESH_PEERS` is currently set to 1, which is a **testnet configuration** suitable for small validator sets (3 nodes). In production, this value must be increased to prevent a single Sybil peer from capturing a validator's entire network view. A production recommendation of `MIN_MESH_PEERS ≥ 3` (or a fraction of the validator set) will be established based on mainnet validator set size.

Peer discovery. Nodes exchange `PeerDiscovery` messages containing their known peer addresses. New peers are added to the connection pool and validated against the validator identity service. The `/api/peers` endpoint exposes the current P2P peer list as the primary field, with legacy HTTP peer information included only when non-empty.

Validator identity verification. The `ValidatorIdentityService` maps P2P peer IDs to validator addresses and BLS public keys. When a new peer connects, the service verifies the peer's claimed identity against the registered validator set. This prevents Sybil attacks at the network layer - only peers corresponding to staked validators contribute to stake visibility calculations for partition detection.

14. Related Work

Rinku draws on and extends several lines of prior work in distributed ledgers, partition-tolerant systems, and proof-carrying architectures. This section positions Rinku honestly within the landscape, crediting the systems that inspired specific design choices and identifying where Rinku's combination of properties is genuinely novel.

14.1 DAG-Based Ledgers

DAG-based transaction structures have been explored extensively as alternatives to linear blockchains:

IOTA (Tangle). IOTA [6] pioneered the use of a DAG for transaction ordering in a distributed ledger, with each transaction approving two prior transactions. The original Tangle design targeted IoT microtransactions and eliminated mining fees. However, IOTA relied on a centralized Coordinator for finality until its Coordicide effort, and the Tangle's tip selection algorithm was vulnerable to parasite chain attacks. IOTA does not provide partition tolerance - the Coordinator is a single point of failure for finality. Rinku shares the DAG structure but differs fundamentally in its consensus mechanism (stake-weighted convergence finality with non-blocking state snapshots) and its explicit partition-tolerant design.

Nano (Block Lattice). Nano [7] uses a block-lattice structure where each account maintains its own chain. This enables asynchronous, feeless transactions with near-instant confirmation via delegated proof-of-stake voting. Nano's design is elegant for simple value transfers but does not support smart contracts or programmable state. Nano does not address network partitions - its Open Representative Voting assumes persistent connectivity to delegates. Rinku's account-nonce model is structurally similar to Nano's per-account chains but embeds transactions in a shared DAG with explicit cross-account ordering via checkpoints.

Sui (Mysticeti). Sui [3] developed the Mysticeti consensus protocol, which achieves sub-round-trip latency by using an uncertified DAG where blocks themselves serve as implicit votes. Rinku's earlier design drew on Mysticeti's insight of sub-round-trip confirmation through parallel voting, but RCC has since evolved to a convergence-based architecture more closely aligned with the Avalanche protocol family [9]. RCC uses explicit `TxConfirmAck` messages for per-transaction convergence (rather than DAG-embedded certificates), and replaces leader-based checkpoint finality with non-blocking state snapshots. Sui does not address partition tolerance - Mysticeti halts when the network cannot reach quorum. See Section 6 for the full RCC architecture.

Avalanche. Avalanche [9] introduced the concept of metastable consensus through repeated sub-sampled polling, achieving probabilistic finality without leaders. RCC draws on this convergence insight but differs in mechanism: RCC uses a single broadcast round where all validators participate (rather than repeated random sub-sampling), achieving deterministic 2/3-stake convergence in a single network round-trip. Avalanche does not provide partition tolerance with deterministic reconciliation.

Other DAG protocols. Hashgraph [8] uses a gossip-about-gossip protocol with virtual voting; Bullshark and Tusk use certified DAGs with structured commit rules. A comprehensive survey of DAG-based consensus is provided by Raikwar et al. [2]. Among these, none provide explicit

partition tolerance with deterministic reconciliation - all assume or require persistent quorum reachability for safety and liveness.

14.2 Partition-Tolerant Distributed Systems

SwarmDAG. SwarmDAG [1] is the direct inspiration for Rinku's partition tolerance design. Tran et al. proposed a partition-tolerant distributed ledger for swarm robotics using a DAG structure that continues operating during network fragmentation. SwarmDAG demonstrated the viability of DAG-based ledgers in intermittently connected environments and introduced the concept of merge-on-reconnect for divergent DAG state. Rinku extends SwarmDAG's approach in several directions: a formalized 5-phase merge protocol with provable deterministic convergence (Section 9), a transaction classification system that governs which operations are safe during partitions (Section 9.8), graduated economic penalties that distinguish intentional abuse from accidental conflicts (Section 12), and a partition budget system that provides deterministic merge-safety guarantees for opted-in accounts.

CRDTs and eventual consistency. Conflict-free Replicated Data Types (CRDTs) guarantee convergence without coordination by restricting operations to commutative, associative, and idempotent structures. Rinku does not use CRDTs for its core financial state because token transfers are inherently non-commutative - order matters and conflicts have economic consequences. However, CRDTs are identified as a natural extension for contract storage types (Section 15.1) where merge-friendly data structures would be safe for unrestricted partition-mode operation.

Bayou and anti-entropy. Bayou [10] pioneered application-level conflict resolution with dependency tracking and tentative/committed state separation. Rinku's three-tier receipt model (tentative, final, reconciliation; Section 9.9) follows a similar conceptual structure, and the distinction between provisional and finalized state maps directly to Bayou's tentative/committed semantics.

14.3 Proof-Carrying and Lightweight Client Approaches

SPV (Simplified Payment Verification). Bitcoin's SPV model [4] proves transaction inclusion via Merkle paths but requires the verifier to trust block headers, which in turn requires following the longest-chain rule. SPV proofs do not prove account state, only transaction inclusion, and the verifier must maintain a header chain or trust a provider. Rinku's self-contained proofs carry the full verification context (Section 4.2), eliminating the need for any ongoing chain-following.

IBC Light Clients. The Inter-Blockchain Communication protocol uses light clients that track validator committees and their signatures. Light clients require an ongoing connection to at least one honest full node to follow validator set evolution. Without this connection, the client's view becomes stale and unverifiable. Rinku's VerifiableObjects are self-contained at the cost of embedding more data per proof (~2KB for an account proof), but eliminate the ongoing connection requirement entirely.

Stateless Ethereum proposals. Ethereum's research into statelessness (Verkle trees, witness-based block execution) aims to reduce full-node storage requirements by embedding state access proofs in blocks. This is a block-producer optimization - clients still need to follow the chain. Rinku's proof-carrying contracts (Section 10.4) target a different use case: enabling application clients that never follow the chain at all, instead verifying individual claims on demand.

14.4 Differentiation

No single existing system combines all five of the properties that Rinku targets. The following table compares across the primary axes:

Property	IOTA	Nano	Sui/Mysticeti	SwarmDAG	Rinku
DAG-based transaction structure	Yes	Yes (block-lattice)	Yes (uncertified DAG)	Yes	Yes
Sub-second transaction finality	No (Coordinator)	Yes (~0.2s)	Yes (~0.4s)	No	Yes (~43ms convergence)
Deterministic partition tolerance	No	No	No (halts)	Yes (basic merge)	Yes (5-phase merge with proofs)
Self-contained offline proofs	No	No	No	No	Yes (VerifiableObjects)
Programmable smart contracts	Yes (ISC)	No	Yes (Move)	No	Yes (WASM)

The individual columns are not novel - DAG structure, fast finality, partition tolerance, proof portability, and smart contracts all exist independently in the literature. Rinku's contribution is the specific combination: a system that provides all five simultaneously, with the partition tolerance and proof portability designed to reinforce each other (VerifiableObjects remain verifiable during and after partitions because they carry their own verification context).

This combination is motivated by the target deployment environments described in Section 1: mesh networks and intermittently connected systems where both partition tolerance and infrastructure-free verification are necessary, not optional.

15. Future Work

15.1 CRDT-Compatible State Types

For contract storage, introduce merge-friendly data types (sets, append-only logs, max counters, OR-maps) that can be safely updated during partitions without conflict. Ideal for social, messaging, and collaborative applications. Would integrate with the transaction classification system (Section 9.8) to automatically determine AP-safety - contracts that exclusively use CRDT-compatible state types could be classified as Safe rather than BoundedSpend.

15.2 Object Ownership Model

Explore single-writer object ownership where owned objects can be processed during partitions without conflict, while shared objects remain CP-only. This model is inspired by Sui's object-centric programming model: if an object has a single owner, mutations by that owner are inherently conflict-free across partitions. The challenge is integrating this with Rinku's account-based (rather than object-based) state model.

15.3 Cross-Chain Proof Composability

Leverage BYOP and VerifiableObjects for cross-chain interoperability. Since VOs are self-contained and carry their own verification data, a Rinku VO could be submitted to a contract on another chain (or vice versa) as a **ProofInput**. The receiving chain would verify the proof's BLS signature against a registered Rinku validator set root. This pattern enables receipt-composable bridges without trusted relayers - the bridge contract verifies the proof's mathematical validity rather than trusting a third party to relay state.

15.4 Contract-Level Merge Hooks

Allow contracts to define custom merge resolution logic for their storage, replacing the default "last-write-wins by weight" rule with application-aware conflict resolution. This would address the pathological contract state corruption risk identified in Section 9.1. A contract could implement a **merge_resolve(key, local_value, remote_value, local_weight, remote_weight) -> value** function that is invoked during Phase 5 of the merge protocol for each conflicting storage key.

15.5 Zero-Knowledge Privacy Layer

An optional ZK-SNARK layer for privacy-preserving transactions is under investigation. The goal is to enable sender obfuscation and private value transfers using Groth16 proofs, with verification integrated into the WASM contract runtime. Key design questions remain open: how

ZK proofs interact with the partition tolerance model (provisional proofs cannot be verified without the proving key, which may not be available in disconnected environments), the tradeoff between proof generation latency and transaction confirmation speed, and whether ZK privacy should be a protocol-level feature or a contract-layer concern. Initial prototyping with circomlib circuits has been completed; production integration is pending resolution of these design questions.

16. Conclusion

Rinku occupies a distinct position in the distributed ledger design space. Rather than making a fixed CAP tradeoff, it dynamically navigates the consistency-availability spectrum based on network conditions: sub-second convergence finality when the network is connected, provisional availability during partitions, and deterministic convergence when partitions heal.

The core contribution is not the mode-switching mechanism itself - dynamically adjusting consistency levels is a well-studied concept in distributed systems. The contribution is the set of protocol mechanisms that make this approach practical for a financial ledger:

1. **The transaction classification system** (Section 9.8) transforms partition tolerance from a binary property (available or not) into a graduated spectrum. Safe operations continue without restriction; bounded-spend operations proceed within configurable risk limits; consensus-critical operations halt until quorum is restored. This gives applications and users explicit control over their partition-mode risk exposure.
2. **The 5-phase merge protocol** (Section 9) provides deterministic reconciliation with formally provable convergence. Integer micro-unit accounting eliminates floating-point nondeterminism. Canonical transaction ordering ensures all nodes compute identical results. The cascade rollback algorithm traces economic dependencies exhaustively, preventing subtle state corruption from partial replays.
3. **Graduated economic deterrence** (Section 12) distinguishes intent. Deliberate double-spending (nonce reuse) is expensive and permanent. Accidental overdrafts receive soft, recoverable penalties. Cascade victims bear no cost. This penalty structure makes rational exploitation unprofitable while avoiding punishing honest users caught in ambiguous situations.
4. **VerifiableObjects** (Section 3) collapse the infrastructure requirements for trust. A `rinku://vo/` URL carries everything needed to verify a claim - no full node, no RPC endpoint, no ongoing network connection. This makes Rinku's proofs inherently portable, shareable, and composable - they can be passed as transaction parameters (BYOP), embedded in QR codes, or verified entirely offline.
5. **Proof-carrying contracts** (Section 10.4) extend this portability to smart contract state. Contracts define which state fragments should be provable; every execution produces a `StatefulReceipt` with Merkle proofs and finality certificates. Applications can be

persistently stateless - rendering verified data from receipts rather than maintaining local state.

Together, these mechanisms create a distributed ledger designed for environments where network partitions are a routine operating condition rather than an exceptional failure. Rinku does not claim to solve the fundamental impossibility results of distributed systems - it claims to make a practically useful navigation of those constraints for the specific domain of decentralized financial infrastructure in mesh-native environments.

Appendices

A. Formal Definitions

Definition 1 (Safety). The Rinku protocol satisfies safety if no two finalized (non-provisional) checkpoints at the same height contain conflicting state roots. Under the honest majority assumption ($>2/3$ stake honest), safety holds because: (a) checkpoint BLS aggregate signatures are verified at reception against the known validator set with a $2/3$ quorum requirement (Section 6.3), so accepting a conflicting checkpoint requires $>1/3$ stake to equivocate (sign two different hashes at the same height); and (b) double-signing is detected by the **ConsensusService** and triggers a 15% stake slash. Note: checkpoints received during snapshot sync of legacy data may lack BLS signatures and are accepted on the basis of `prev_hash` chain linkage only; full BLS enforcement applies to all real-time checkpoint announcements.

Definition 2 (Liveness - Normal Mode). The protocol satisfies liveness in normal mode if every submitted valid transaction is eventually included in a finalized checkpoint, assuming $>2/3$ stake is reachable and the leader election mechanism produces a live leader within bounded time. The leader fallback mechanism (Section 6.4) ensures liveness even if the primary leader is offline.

Definition 3 (Liveness - Partition Mode). The protocol satisfies partition liveness if every submitted valid transaction classified as Safe or BoundedSpend (within budget) is included in a provisional checkpoint within bounded time, assuming the local partition contains $\geq 1/3$ total stake. CpOnly transactions do not satisfy liveness during partition by design.

Definition 4 (Convergence). The merge protocol satisfies convergence if, given the same set of transactions from both partitions and the same fork-point state, every node independently computes identical post-merge state. This follows from: (a) integer micro-unit accounting eliminating floating-point nondeterminism, (b) canonical ordering producing a strict total order,

and (c) the cascade rollback algorithm being a deterministic function of its inputs with provable termination.

Definition 5 (Cascade Termination). Let S_0 be the initial set of surviving transactions (after conflict resolution). The cascade rollback algorithm produces a sequence $S_0 \supseteq S_1 \supseteq S_2 \supseteq \dots$ that converges in at most $|S_0|$ iterations. Proof: each iteration either removes at least one transaction ($|S_{i+1}| < |S_i|$) or produces no new rollbacks ($S_{i+1} = S_i$, and the algorithm terminates). Since $|S_i| \geq 0$ and decreases by at least 1 per non-terminal iteration, the algorithm terminates in at most $|S_0|$ steps.

B. Parameter Reference

Parameter	Value	Section	Description
MAX_SUPPLY	30,000,000 RKU	11.1	Hard token supply cap
GENESIS_ALLOCATION	6,000,000 RKU	11.1	Initial token distribution
HALVING_INTERVAL	3,150,000 checkpoints	11.1	Reward halving period
MIN_REWARD_FLOOR	0.122887 RKU	11.1	Minimum checkpoint reward
CHECKPOINT_INTERVAL	~10 seconds	6.2	Target time between checkpoints
CONVERGENCE_QUORUM	2/3 active stake ($\approx 66.67\%$)	6.1	Convergence finality threshold
SNAPSHOT_QUORUM	Single proposer (no quorum)	6.2	Snapshot creation (proposer-signed)
SUPER_MAJORITY	75% total stake	6.3	Higher-security operations
LEADER_TIMEOUT	45 seconds	6.4	Fallback leader election trigger
PROPAGATION_GRACE	5,000 ms	6.2	Transaction propagation window

Parameter	Value	Section	Description
MIN_STAKE	100 RKU	11.4	Minimum validator stake
UNSTAKE_COOLDOWN	24 hours	11.4	Stake withdrawal delay
UNBONDING_PERIOD	14 days	11.4	Slashing vulnerability window
MIN_GAS_PRICE	0.001 RKU	11.6	Gas price floor
MAX_GAS_PRICE	10.0 RKU	11.6	Gas price ceiling
GAS_TARGET	150 tx / 15s	11.6	Target transaction throughput
GAS_ADJUSTMENT	12.5%	11.6	Per-period price adjustment
BURN_CEILING	30%	11.7	Maximum fee burn percentage
MAX_SAMPLED_TIPS	16	7.2	Maximum parent references per tx
PARTITION_THRESHOLD	66.66% visible stake	8.1	Partition detection threshold
T_CONF	30 seconds	8.1	Partition confirmation timeout
T_RECOVERY	10 seconds	8.1	Quorum recovery window
PROVISIONAL_FLOOR	33.33% total stake	8.3	Minimum stake for provisional checkpoints
WEIGHT_PROXIMITY	1.5x	9.4	Weight tiebreaker proximity threshold
NONCE_REUSE_BALANCE_PENALTY	10%	12.1	Balance confiscation for double-spend

Parameter	Value	Section	Description
NONCE_REUSE_STAKE_SLASH	100%	12.1	Stake slash for double-spend
NONCE_REUSE_REPUTATION	0.50 permanent	12.1	Permanent reputation penalty
OVERDRAFT_REPUTATION	0.10 decaying	12.1	Recoverable reputation penalty
REPUTATION_DECAY_PERIOD	100 checkpoints	12.3	Linear decay window (~16 min)
DOUBLE_SIGN_SLASH	15%	12.1	Stake slash for checkpoint equivocation
IDLE_TIMEOUT	600 seconds	13.3	P2P connection idle timeout
MIN_MESH_PEERS	1 (testnet ; production TBD)	13.3	Minimum gossipsub mesh size
MAX_PROPAGATION_BATCH	100	13.1	Transaction propagation batch size
WASM_MAX_PAGES	256 (16 MB)	10.1	Contract memory limit
WASM_DEFAULT_FUEL	10,000,000	10.2	Default instruction fuel budget
MICRO_UNIT_SCALE	10 ⁸	11.8	Micro-units per RKU

C. Benchmarks (Preliminary Testnet Data)

The following benchmarks were collected on a 3-validator Fly.io testnet deployment and represent preliminary testnet performance. Production benchmarks with larger validator sets and realistic network conditions are pending.

C.1 Throughput

TODO

C.2 Convergence Finality Latency

Percentile	Latency
p50	43–44 ms
p95	~200 ms
p99	~500 ms
Min	~22 ms

Measured as time from submission to convergence confirmation. **Methodology note:** only confirmed samples are included; transactions that did not achieve convergence within 10s are excluded. The exclusion rate will be reported in production benchmarks. The p50 of ~43ms demonstrates sub-second finality for the majority of transactions via convergence consensus.

C.3 Snapshot Latency

Percentile	Latency
p50	~10,300 ms
p95	~10,400 ms
p99	~15,400 ms

Snapshot latency measures the time from transaction submission to inclusion in a state snapshot. This is dominated by the snapshot interval (~5-10s). Note that transactions achieve finality much earlier via convergence (Section 6.1, Appendix C.2); snapshot inclusion provides the anchoring point for proof generation and DAG pruning, not finality itself.

C.4 Proof Generation & Size

Proof Type	Generation Time	Response Size	Success Rate
Account proof (Merkle inclusion)	21 ms	1,953 B	100% (5/5)
Transaction proof	26 ms	1,841 B	100% (5/5)
Self-contained proof (VO URL)	21 ms	1,703 B	100% (5/5)

Proof Type	Generation Time	Response Size	Success Rate
Batch proof (multi-receipt)	22 ms	In development	In development

All individual proof types are generated in under 30ms. Self-contained VerifiableObject URLs encode at ~1.7KB, enabling URL-portable verification without external state. Account proofs include the Sparse Merkle Trie inclusion path and weigh ~2KB. **BatchProof** aggregation (shared checkpoint context and Merkle multiproof across multiple receipts) is implemented at the type level but end-to-end benchmarks including response size and multi-receipt verification are pending completion.

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